EXP 2:

Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

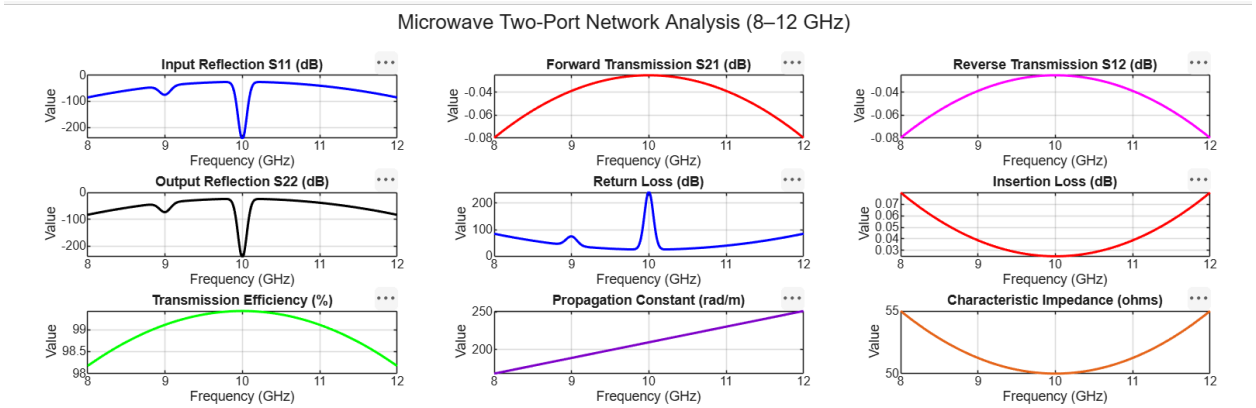
CODE:

```
clc;
clear;
close all;
% Input Parameters
f = 8:0.01:12;          % Frequency in GHz
Z0 = 50;                % Characteristic impedance in ohms
c = 3e8;                % Speed of light in m/s
L = 50e-3;              % Network length in metres
% Convert frequency to Hz
f_Hz = f * 1e9;
% Free-space wavelength and propagation constant (TEM)
lambda0 = c ./ f_Hz;
beta = 2*pi ./ lambda0;
% Illustrative S-parameter model (dB)
S11_dB = -25 - 15*(f-10).^2 ...
  - 35*exp(-((f-9)/0.1).^2) ...
  - 220*exp(-((f-10)/0.08).^2);
S11_dB = max(S11_dB, -240);
S22_dB = max(S11_dB + 2, -240);
S21_dB = -0.025 - 0.055*((f-10)/2).^2;
S12_dB = S21_dB;
% Convert S-parameters from dB to linear magnitude
S11 = 10.^(S11_dB/20);
S22 = 10.^(S22_dB/20);
S21 = 10.^(S21_dB/20);
S12 = 10.^(S12_dB/20);
% Return Loss and Insertion Loss
RL = -20*log10(S11);
IL = -20*log10(S21);
% Transmission efficiency
Eff = (S21.^2)*100;
% Illustrative characteristic impedance model
Zc = Z0 + 1.25*(f-10).^2;
% Phase shift across the network
Phase_shift = beta * L;
```

```
Phase_deg = Phase_shift * 180/pi;
% Find optimum operating frequency
[~, idx_min] = min(S11);
[~, idx_max] = max(S21);
% Display results at 10 GHz
[~, idx10] = min(abs(f-10));
fprintf('\nMICROWAVE TWO-PORT NETWORK RESULTS\n');
fprintf('--------------------------------\n');
fprintf('Frequency            = %.2f GHz\n', f(idx10));
fprintf('S11                  = %.3f dB\n', S11_dB(idx10));
fprintf('S21                  = %.3f dB\n', S21_dB(idx10));
fprintf('S12                  = %.3f dB\n', S12_dB(idx10));
fprintf('S22                  = %.3f dB\n', S22_dB(idx10));
fprintf('Return Loss          = %.3f dB\n', RL(idx10));
fprintf('Insertion Loss       = %.3f dB\n', IL(idx10));
fprintf('Transmission Efficiency= %.3f %%\n', Eff(idx10));
fprintf('Propagation Constant   = %.3f rad/m\n', beta(idx10));
fprintf('Characteristic Impedance = %.3f ohms\n', Zc(idx10));
fprintf('Phase Shift          = %.3f degrees\n', Phase_deg(idx10));
fprintf('\nOPTIMUM FREQUENCY RESULTS\n');
fprintf('------------------------\n');
fprintf('Minimum S11 frequency  = %.2f GHz\n', f(idx_min));
fprintf('Maximum S21 frequency  = %.2f GHz\n', f(idx_max));
% Plot results
figure('Color','w');
Y = {S11_dB, S21_dB, S12_dB, S22_dB, ...
   RL, IL, Eff, beta, Zc};
T = {'Input Reflection S11 (dB)', ...
   'Forward Transmission S21 (dB)', ...
   'Reverse Transmission S12 (dB)', ...
   'Output Reflection S22 (dB)', ...
   'Return Loss (dB)', ...
   'Insertion Loss (dB)', ...
   'Transmission Efficiency (%)', ...
   'Propagation Constant (rad/m)', ...
   'Characteristic Impedance (ohms)'};
C = {'b','r','m','k','b','r','g',[0.5 0 0.8],[0.9 0.4 0.1]};
for k = 1:9
  subplot(3,3,k);
  plot(f,Y{k},'Color',C{k},'LineWidth',1.5);
```

```
    title(T{k});
    xlabel('Frequency (GHz)');
    ylabel('Value');
    grid on;
    xlim([8 12]);
end
sgtitle('Microwave Two-Port Network Analysis (8–12 GHz)');
```

OUTPUT:



VERIFICATION:

Exp-2

$f_0 = 10\text{GHz}$ $\quad Z_0 = Z_L = 50\Omega$

$L = 50\text{mm}$

propagation constant

$$\beta = 2\pi f/c = \frac{2\pi(10\times10^9)}{3\times10^8}$$

$$\boxed{\beta = 209.44 \text{ rad/m}}$$

phase shift $\quad \phi = \beta L = 209.44\,(0.05) = 10.472 \text{ rad}$

$$\boxed{\phi = 600^\circ}$$

$S_{11} = -30\text{dB}$

$$|S_{11}| = 10^{-30/20} = 0.03162$$

Return loss $\quad R_L = -20\log_{10}|S_{11}|$

$$= -20\log_{10}(0.03162)$$

$$\boxed{R_L = 30\text{dB}}$$

$$|S_{21}| = 10^{-1/20} = 0.89125$$

$$IL = -20\log|S_{21}| = -20\log(0.89125)$$

$$\boxed{IL = 1\text{dB}}$$

efficiency $\quad \eta = |S_{21}|^2 \times 100\%$

$$= (0.89125)^2 \times 100\%$$

$$\boxed{\eta = 79.43\%}$$